

Dynamic Evolutionary Optimization Algorithm Switching in AutoML Systems

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Guide Approval



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Namah Shivaya

Approved the PPT for panel presentation.

with regards

Thangavelu S



Reply



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Review Feedback & Actions Taken

Feedback	Action Taken
How will you measure stagnation?	No fitness improvement for k consecutive iterations; monitored at both Tier 1 and Tier 2
What are your metrics for switching?	5-metric composite score: Accuracy, Speed, Efficiency, Diversity, Trend (weighted sum)
Algorithms are sequential — what if they cooperate?	Redesigned to tiered island model — 4 algorithms run in parallel; T2 islands share elites
How do you choose the starting algorithm?	Tier-1 warm-up: all 4 compete for k iterations; top 2 promoted to Tier 2 by composite score
What is meta-training? How to use it?	UCB1 bandit acts as meta-learning layer — learns and selects best algorithm during stagnation
Final system is design not clearly defined	Added complete architecture diagram, end-to-end workflow, and edge case handling

1. Literature Survey & Research Direction

Survey Scope and Key Findings

Survey Scope:

- Focused on:
 - Model selection + hyperparameter optimization
 - Optimization efficiency and convergence
 - Adaptability of search algorithms

Key Findings:

- **XAutoLM** [1]: Meta-learning improves search, but uses fixed algorithm
- **SHSR** [2]: Reduces search space, no runtime adaptation
- **OnMAR** [3]: Online learning, but single optimizer
- **MsMA** [4]: Handles stagnation, no algorithm switching

1. Literature Survey & Research Direction (Contd.)

Research Gap and Proposed Direction

Research Gap Identified [1, 2, 3, 4, 5, 6]:

- **No Dynamic Algorithm Switching:** Single fixed optimizer used throughout
- **Limited Adaptability:** Cannot adjust to changing search behaviour
- **Disconnected Stagnation Handling:** Not integrated with decision-making
- **Poor Exploration-Exploitation Balance:** Leads to premature convergence

Proposed Direction:

- **A hybrid framework:**
 - **Tiered Competition (40-60 split):** 4 algorithms (GA, PSO, DE, CMA-ES) compete continuously in Tier 1; top 2 promoted to Tier 2 for real optimization
 - **Island-Based Co-operative Co-evolution:** Tier-2 algorithms operate as islands with periodic elite sharing for cross-pollination
 - **UCB1 Bandit for Stagnation:** Balances exploitation (use what works) and exploration (try underused algorithms) when performance plateaus
 - **Dual-Tier Stagnation Handling:** Monitors and responds to stagnation at both Tier 1 and Tier 2

2. Originality of Work

Rubric: Novel Idea vs. Extension

What Existing Systems Do

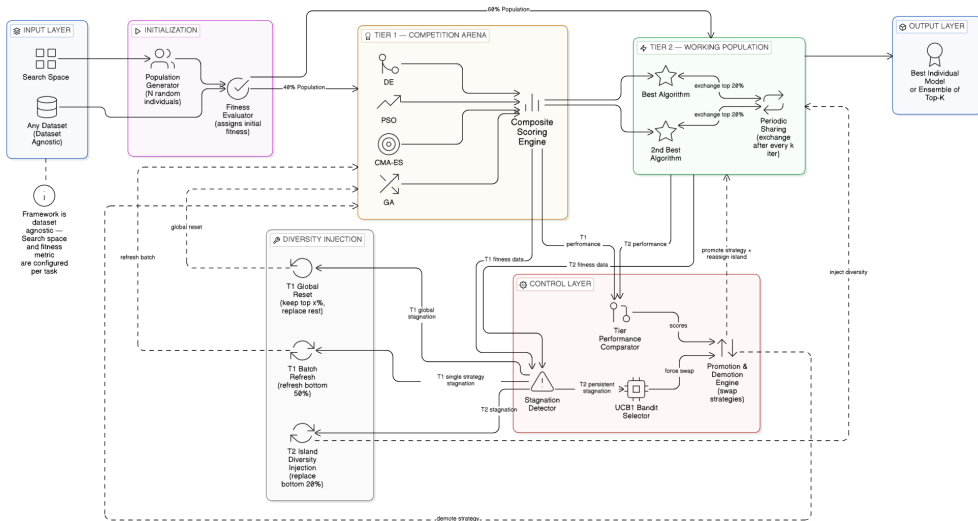
- Use a **single fixed optimizer** for the entire search (no switching)
- If they use multiple algorithms, they run them **independently** with no communication
- Stagnation is either ignored or handled by restarting the search
- Poorly performing algorithms **waste resources** with no mechanism to reallocate

What Our System Does Differently

- **No algorithm is ever eliminated** — all 4 algorithms run continuously in Tier 1, always available for promotion
- **Algorithms earn resources meritocratically** — only the top 2 get the larger Tier-2 population to work with
- **Knowledge flows between algorithms** — periodic sharing cross-pollinates solutions across islands
- **Stagnation triggers intelligent response** — UCB1 bandit selects the next algorithm to try, not random switching
- **Both tiers are monitored** — underperforming T1 algorithms get batch refresh; T2 islands get diversity injection

3. System Architecture Overview

End-to-End Dynamic Algorithm Switching Flow



4. Hybrid Model: Tiered Islands with Bandit-Based Promotion

Core innovation of the project

Tier 1 — Competition Arena (40%):

- All 4 algorithms (GA, PSO, DE, CMA-ES) compete on small batches
- **Always-on** — no algorithm is ever permanently eliminated
- Composite scores computed every k iterations
- Stagnating algorithms get batch refresh + temporary size reduction

Tier 2 — Working Population (60%):

- Top 2 algorithms from Tier 1 get promoted here as **islands**
- Periodic sharing: top 20% individuals exchanged every k iterations
- Cross-pollination ensures genetic diversity

Promotion Mechanism:

- If a Tier-1 algorithm's composite score $>$ a Tier-2 algorithm's score $\Rightarrow$ **promote and replace**
- Grace period after promotion (no challenges for $2k$ iterations)
- Demoted algorithm returns to Tier 1 — can re-compete

Stagnation Handling (both tiers):

- **T2:** Inject 20% random $\Rightarrow$ if persists, UCB1 bandit switch
- **T1 single:** Refresh 50% batch + temporarily shrink allocation
- **T1 global:** Keep top $x\%$ best per island, refresh rest, re-warm-up

5. Composite Scoring & UCB1 Bandit

How algorithms are evaluated and selected

Composite Score (per algorithm):

Each algorithm is scored on 5 metrics. Each metric is first **normalized** (min-max across all algorithms so scores range 0–1), then multiplied by its weight. The final score is the **weighted sum** of all normalized metrics.

Metric	Weight
Accuracy / Fitness	0.30
Convergence Speed	0.25
Computational Efficiency	0.15
Population Diversity	0.15
Performance Trend	0.15

Higher composite score = better-performing algorithm.
Used for promotion decisions every k iterations.

UCB1 Bandit (for stagnation response):

When a Tier-2 island stagnates persistently, the system uses UCB1 to pick which algorithm to try next. The UCB1 score for each algorithm combines two parts:

- 1 **Exploitation:** The algorithm's average past reward — how well it has performed historically
- 2 **Exploration bonus:** A bonus that grows larger for algorithms that have been used *fewer* times — ensures underused algorithms get a fair chance

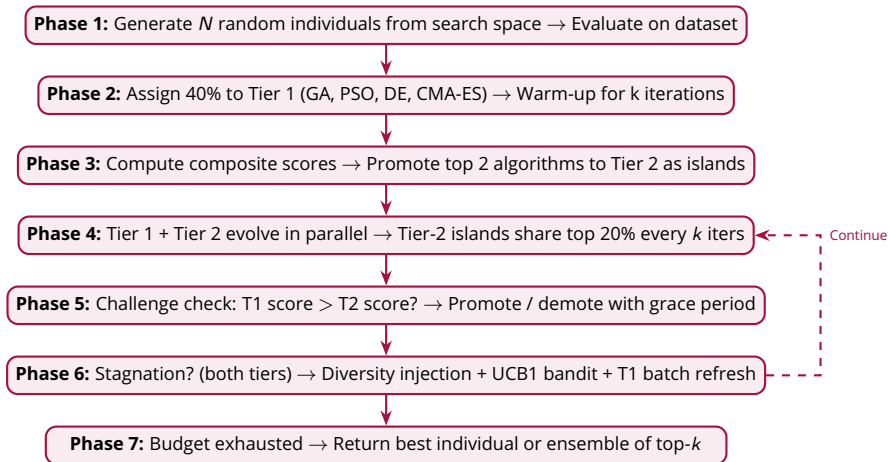
The algorithm with the **highest combined score** (reward + exploration bonus) is selected.

Key Insight

An algorithm with low average reward but very few uses can still be selected due to its high exploration bonus. This prevents the system from permanently ignoring any algorithm.

6. End-to-End Workflow

Single-run execution flow



7. Edge Cases & Robustness

How the system handles failure scenarios

Edge Case	Solution	Safeguard
Single T1 algorithm stagnates	Refresh bottom 50% individuals; if persists temporarily shrink batch and redistribute to best performing algorithm	Resource efficiency
All T1 algorithms stagnate	Keep top x% individuals per island -> replace rest with random -> re-warm-up	Global refresh
Promoted algorithm fails on larger population	5-iterations grace period; if no gain, rollback to the state before the switch	Snapshot & restore
Oscillating promotions ($A \leftrightarrow B$)	Cooldown of $2k$ iterations; require margin $\delta=0.05$ to replace	Stability guarantee
T2 islands converge to same region	Reduce sharing frequency; inject 10% random individuals into weaker island	Diversity preservation
Full population collapse	Soft reset: keep top 30% individuals, replace 70% with random, re-warm-up	Emergency recovery

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Thank You